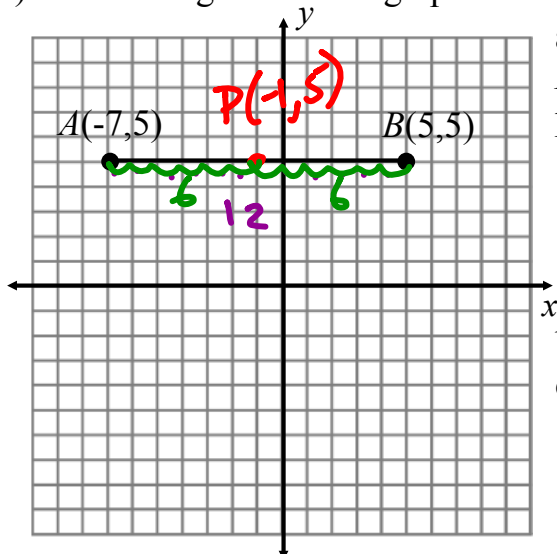


Students will calculate midpoints of line segments.

Do Now

1) The line segment $\overline{AB}$ is graphed below.



a. Determine the location of point P that splits $\overline{AB}$ into two equal parts.

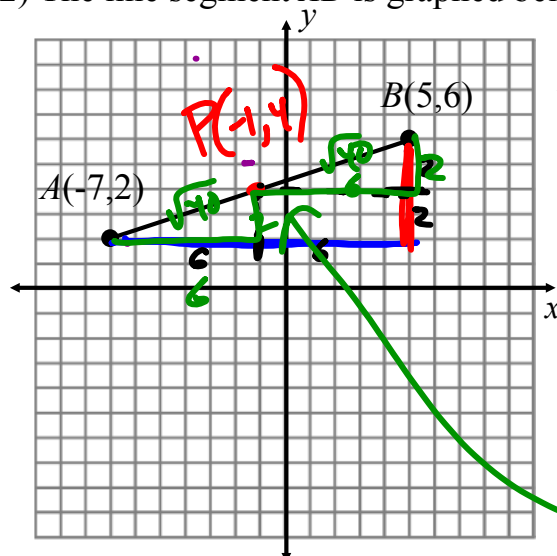
b. How do you know that P is directly in the middle of $\overline{AB}$?

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Do Next

$$\left(\frac{-7+5}{2}, \frac{2+6}{2} \right)$$

2) The line segment $\overline{AB}$ is graphed below.

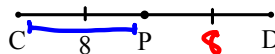
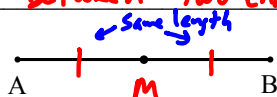


a. Determine the location of point P that splits $\overline{AB}$ into two equal parts.

b. Explain how you know that P splits $\overline{AB}$ into two equal parts.

Midpoints & Bisectors

A **midpoint** of a line segment is a point (x,y) that lies exactly in between two endpoints splitting the segment in 2 equal parts.



$$CD = 8 + 8 = 16$$

A **segment bisector** is any point, line, ray, segment, or plane that cuts a segment into 2 congruent parts.

Point P is a segment bisector because it cuts $\overline{EF}$ into two equal parts.



Line l is a segment bisector because it cuts $\overline{EF}$ into two equal parts.

Practice
In each question, identify the segment bisector and the length indicated.

1.	2.	3.
Segment Bisector: Point M	Segment Bisector: line m	Segment Bisector: Point M
$13 + 13 = 26$	$ \begin{aligned} x + 15 &= 7x - 3 \\ -x &\quad -x \\ 15 &= 6x - 3 \\ +3 &\quad +3 \\ 18 &= 6x \\ \frac{18}{6} &= \frac{6x}{6} \\ 3 &= x \end{aligned} $	$ \begin{aligned} 6x &= 2x + 4 \\ &\vdots \\ x &= 1 \\ EM &= 6(1) = 6 \\ MF &= 2(1) + 4 = 6 \end{aligned} $
AB = 26	EM = $(3) + 15$ = 18	EF = $6 + 6$ = 12

How to find a Midpoint Graphically

Find the midpoint of $(-4, 3)$ and $(8, 7)$.

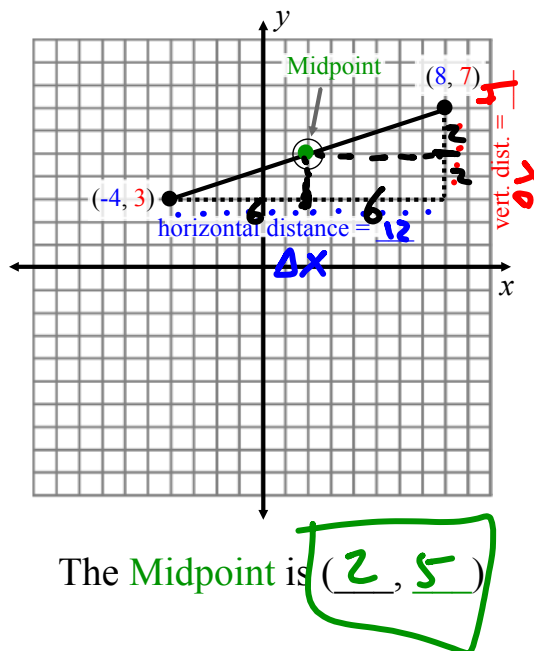
Step 1: Plot the two points and connect them

Step 2: Find the **horizontal distance** between the two points

Step 3: Find the **vertical distance** between the two points

Step 4: Divide the horizontal distance and the vertical distance by 2

Step 5: Determine the point created by connecting the halfway points of the vertical and horizontal distances



How to find a Midpoint Algebraically

Find the midpoint of the segment with endpoints $(-4, 3)$ and $(8, 7)$.

Step 1: Label the coordinates of the points (x_1, y_1) and (x_2, y_2)

Step 2: Add the **x-coordinates**, then divide by 2.

This is the **x-coordinate** of the Midpoint.

Step 3: Add the **y-coordinates**, then divide by 2.

This is the **y-coordinate** of the Midpoint.

Step 4: Write your answer as a point

Step 1: x_1, y_1 and x_2, y_2
 $(-4, 3)$ and $(8, 7)$

Step 2: $\frac{-4 + 8}{2} = \frac{4}{2} = 2$

Step 3: $\frac{3 + 7}{2} = \frac{10}{2} = 5$

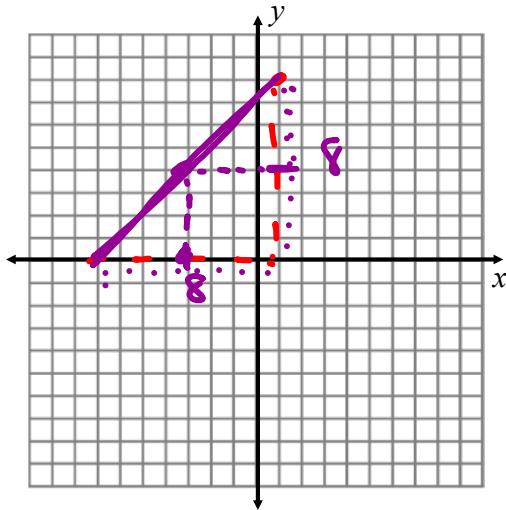
Step 4: The Midpoint is $(2, 5)$

Midpoint Formula:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Average of x-coordinates Avg. of y-coordinates

- 4) Find the midpoint of the segment with endpoints (1, 8) and (-7, 0)



The Midpoint is $(-3, 4)$

- 5) Find the midpoint of the segment with endpoints (2, 5) and (6, -9)

$$\frac{2+6}{2} = \frac{8}{2} = 4$$

$$\frac{5+(-9)}{2} = \frac{-4}{2} = -2$$

$$(4, -2)$$

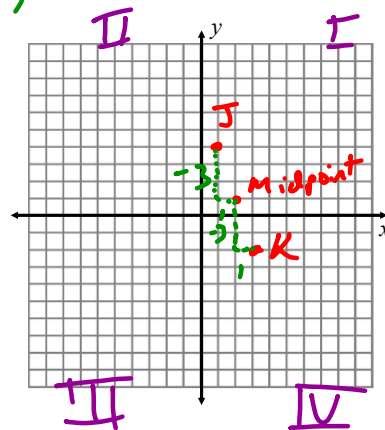
The Midpoint is $(4, -2)$

In each question below, the midpoint is represented by M . Find the missing midpoint or endpoint. The use of the graph is optional.

- 6) $\overline{RS}$: $R(1, -3)$ M _____ $S(4, 2)$
- 7) $\overline{JK}$: $J(1, 4)$ $M(2, 1)$ $K(3, -2)$
- 8) $\overline{RS}$: $R(-8, 7)$ M _____ $S(5, 1)$
- 9) $\overline{JK}$: $J(4, 1)$ $M(2, -3)$ K _____

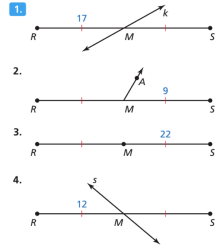
$$\frac{1+x}{2} = 2$$

$$\frac{4+y}{2} = 1$$

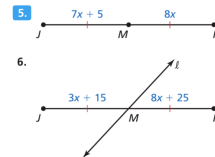


Assignment 1.6: Segment Bisectors & Midpoints

For questions 1 - 4: Identify the segment bisector of $\overline{RS}$ then find RS .



For questions 5 - 6: Identify the segment bisector of $\overline{JK}$ then find JK .

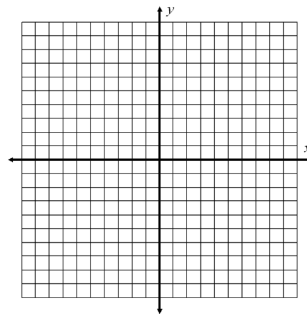


7. The endpoints of $\overline{CD}$ are given. Find the coordinates of the midpoint, M . [The use of the graph is optional]

a. $C(3, -5)$ and $D(7, 9)$

b. $C(-2, 0)$ and $D(4, 9)$

c. $C(-8, -6)$ and $D(-4, 10)$



8. The midpoint M and one endpoint of $\overline{GH}$ are given. Find the coordinates of the other endpoint. [The use of the graph is optional]

a. $H(-3, 7)$ and $M(-2, 5)$

b. $G(5, -6)$ and $M(4, 3)$

c. $G(-4, 1)$ and $M(-\frac{13}{2}, -6)$

